Trust-Weighted Multi-Agent Co-Learning for Portfolio Optimization

Mathematics for Finance — Project Proposal

Motivation

Portfolio-managing agents operating in shared markets can observe peer decisions and potentially improve their own strategies by selectively incorporating peer signals. We fix a trust mechanism and check which learning strategies beneficial, stable, and robust.

Problem Formulation

M agents each manage a portfolio over N risky assets in a discrete-time regime-switching market. Each agent i holds a trust vector τ_i^t over peers, updated via a standard exponential moving average on realized signal quality:

$$\tau_{ij}^{t+1} = (1-\alpha)\tau_{ij}^t + \alpha \cdot \exp(-\lambda|\hat{s}_j^t - r^t|_1)$$

where $\hat{s}_j^t$ is agent j's predicted return and r^t is the realized return. Trust is initialized uniformly and the aggregated peer signal used in decision-making is the trust-normalized weighted mean of peer predictions. Each agent maximizes its own risk-adjusted return: $E[\Sigma_t (r_{i,t} - \gamma/2 \cdot \mathrm{Var}(r_{i,t}))]$.

Learning Strategies

- **S1:** Independent Q-Learning (Baseline)
- **S2:** Trust-Weighted Policy Imitation: agent blends own action with trust-weighted peer actions, $\hat{a}_i = (1-\beta)a_i + \beta\Sigma_j\tau_{ij}a_j$

Research Questions

RQ	Question	Metric
1	Does trust converge, and how quickly?	Trust entropy $H(\tau_i^t)$ over time
2	Does cooperation via trust improve portfolio performance?	Sharpe ratio and cumulative return vs. S1 baseline
3	Is trust robust to market regime shifts?	Trust drift and recovery time post-regime switch
4	Does asymmetric trust ($\tau_{ij} \neq \tau_{ji}$) emerge and persist?	Asymmetry index $\Sigma_j\lvert\tau_{ij}-\tau_{ji}\rvert/M$; correlation with performance gap

Implementation

- The market environment will be implemented as a custom Gymnasium simulator.
- Agents will be implemented in PyTorch.
- The three strategies will be evaluated across two conditions: homogeneous agents (same information set) and heterogeneous agents (asset specialization).
- We will also explore regime-switching returns in various regime shift scenarios